

Improved Quantum Circuits for Information Set Decoding with Application to Code-Based Cryptography

Hyunji Kim Kyungbae Jang Hwajeong Seo

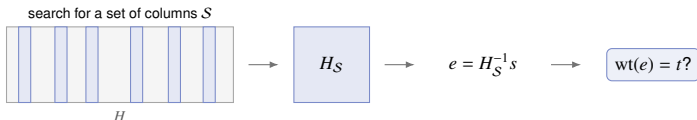
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Motivation

- **Code-based cryptography is one of the main families of post-quantum cryptography.**
- **Its security relies on the syndrome decoding problem:** recovering a vector e of low Hamming weight t from $He = s$.
- **Information set decoding is the best known attack on that problem, and Grover's search accelerates it.**

In this work, we present an optimized quantum ISD circuit for HQC and Classic McEliece.

Information Set Decoding (ISD)



- Parity-check matrix (H) and syndrome (s):** each column corresponds to a possible error position. The syndrome is determined by the columns at the actual error positions.
- Search target:** Each column of H represents a possible error position. Given a candidate set S of r column indices, the submatrix H_S consists of the columns of H selected by S . ISD searches for a set S that contains all error positions.
- Linear-system solution:** For an invertible H_S , solving the linear system yields $e = H_S^{-1}s$, which is the candidate error vector on the selected positions.
 The error vector e indicates which positions contain errors, so we need to recover e to solve SDP.
 Two representative approaches are Gauss–Jordan elimination and Wiedemann-based sparse inversion.
- Hamming-weight test:** To determine whether the required error vector has been recovered, ISD checks whether the resulting vector e has Hamming weight t .

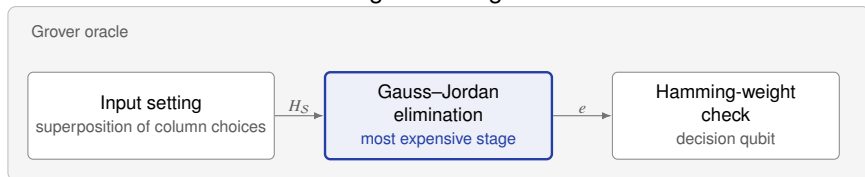
Quantum ISD (QISD)

- **Grover search:** QISD applies Grover search to accelerate ISD's search for S .
- **Oracle:** For each candidate set S , the oracle solves the linear system and checks the Hamming weight of the solution.
- **Prange's ISD:** the representative variant. In this work, we adopt Prange ISD.

Advanced variants (more than 40 ISD variants) reduce the number of searches but raise the cost of one oracle call; no reported circuit translation gives a clear concrete advantage.

Structure of the QISD Oracle

The QISD oracle has the following three stages.



- **Input setting**: prepares a superposition over candidate sets S and gathers the selected columns into H_S .
- **Gauss-Jordan elimination**: is used to solve the linear system using H_S and s .
- **Hamming-weight check**: determines the value of the decision qubit based on whether the solution has Hamming weight t .

Related Quantum ISD Circuits

Prior studies provide the QISD circuit with an optimized Gauss–Jordan elimination, or a sparse-inversion alternative.

Source	Circuit	Relation to this work
P ⁺ [25] (2023)	First complete QISD circuit, Gauss-Jordan elimination	Input setting and iteration counts adopted in this work.
J ⁺ [15] (2024)	Depth-optimized parallel Gauss–Jordan elimination	Parallel elimination structure adopted and further optimized in this work.
Ch ⁺ [8] (2024)	Wiedemann sparse inversion, qubit-minimal	Opposite end of the time–space trade-off, with fewer qubits and more gates.

We improve the elimination circuit in the QISD oracle, reducing the qubit and gate counts while keeping the depth comparable to J+ [15].

Input Setting

Table 1: Quantum resources for the input setting, re-expressed in closed form.

Stage	#Qubit	# T	T -depth
Dicke state [21]	N	$149(4Nr - 4r^2 - 2N + 1)$	–
Sorting network [25]	$Nr + N + \frac{N\ell(1 + \ell)}{4}$	$\frac{N\ell(1 + \ell)}{4}(7r + 7)$	$2\ell(1 + \ell) + 4r$

N : code length, $r = N - k$, and $\ell = \lceil \log_2 N \rceil$.

- **Dicke state** $|D_r^N\rangle$: an equal superposition over all N -bit strings of Hamming weight r , where each 1 indicates a selected column.
- **Sorting network**: moves the marked columns to the left, so the later stages act on a contiguous submatrix.

Previous Gauss-Jordan elimination

	col0	col1	col2	col3	col4	col5	col6	col7	s
row0	.	.	.	.	.	.	.	.	.
row1	.	.	.	.	.	.	.	.	.
row2	.	.	P	.	.	.	.	.	.
row3	L	L	.	R	R	R	R	R	S
row4	L	L	.	R	R	R	R	R	S
row5	L	L	.	R	R	R	R	R	S
row6	L	L	.	R	R	R	R	R	S
row7	L	L	.	R	R	R	R	R	S

P: pivot *L*: lower-left *R*: lower-right *S*: syndrome

- Matrix update.** The syndrome $s = He$ implicitly contains the error vector. Elimination is applied to $[H_S \mid s]$ from the top row to the bottom row to recover error vector.
- L , R and S are all updated.** The pivot row is added to every lower row with a 1 in the pivot column, and the update covers the left elements of the pivot (L), its right elements (R) and the syndrome elements (S).
- Parallelization with pivot fan-out.** The rows (L , R , S) are updated in parallel using the copied pivot data.

One copy per update is required, so the copies for the L and R make this step the qubit peak of the oracle.

Optimized Gauss-Jordan elimination by us

	col0	col1	col2	col3	col4	col5	col6	col7	s
row0	.	.	.	.	.	.	.	.	.
row1	.	.	.	.	.	.	.	.	.
row2	.	.	P	.	.	.	.	.	.
row3	.	.	.	R	R	R	R	R	S
row4	.	.	.	R	R	R	R	R	S
row5	.	.	.	R	R	R	R	R	S
row6	.	.	.	R	R	R	R	R	S
row7	.	.	.	R	R	R	R	R	S

P: pivot *L*: lower-left *R*: lower-right *S*: syndrome

- **Updating *R* and *S* only, to avoid the redundant updates.** *R* contains the next pivot and has to be updated, whereas *L* is not used directly by the operations of the next pivots.
- **Back-substitution.** Forward elimination does not yet produce the complete error vector, so an additional operation on the syndrome is required to obtain the correct result.
- **Lower quantum resources.** Omitting the parallel updates on *L* removes the ancillas and the Toffoli gates they required.

Optimized Gauss-Jordan elimination by us

Algorithm 1: Forward-elimination phase of Algorithm 1 (lines 1-9).

Require: $[H \mid s]$ with $H \in \mathbb{F}_2^{r \times r}$ (H_{ik} is row i , column k)

```

1: for  $j = 0$  to  $r - 1$  do                                     ▶ forward elimination
2:    $ok_j \leftarrow \text{PivotFix}(H, j)$                              ▶ set the pivot  $H_{jj}$  to 1; record success
3:   for  $i = j + 1$  to  $r - 1$  do
4:     for  $k = j + 1$  to  $r - 1$  do                                     ▶ right block  $R$  only
5:        $H_{i,k} \leftarrow H_{i,k} \oplus H_{i,j}H_{j,k}$ 
6:     end for
7:      $s_i \leftarrow s_i \oplus H_{i,j}s_j$                              ▶ syndrome  $S$ 
8:   end for
9: end for
  
```

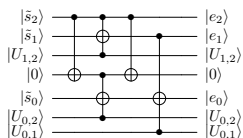
- **The pivot is set to 1 (line 2)**, so that it can eliminate the entries below it.
- **The AND of the pivot-column entry ($H_{i,j}$, below the pivot) and the pivot-row entry ($H_{j,k}$, right of the pivot) is added to $H_{i,k}$ of R by a Toffoli gate (line 5).**
- **The syndrome s_i is updated in the same way, by applying a Toffoli gate to $H_{i,j}$ and s_j (line 7).**
- **At each pivot, the updates of R and S are all executed in parallel (lines 3–8)**, the pivot column and row being copied onto ancillas that are uncomputed and reused at the next pivot.

Optimized Gauss-Jordan elimination by us

Algorithm 1: Syndrome back-substitution (lines 10-14).

```

10: for  $j = r - 1$  downto 1 do
11:   for  $i = 0$  to  $j - 1$  do
12:      $s_i \leftarrow s_i \oplus H_{ij}s_j$ 
13:   end for
14: end for
  
```



Parallel back-substitution at the gate level for $r = 3$

- Forward elimination updates only the rows below each pivot, so the last row is processed by all preceding pivots. Therefore, the last syndrome bit is guaranteed to equal the last error bit.
- Starting from the last syndrome bit, Toffoli gates use each recovered solution bit to update the syndrome bits above. This recovers the remaining bits upward and converts the syndrome register into the solution.
- All updates of inner loop ($i < j$, line 11) is performed in parallel.
- All ancilla qubits from the forward-elimination step are reused.

This reduces the qubit and gate counts with a comparable circuit depth.

Optimized Gauss-Jordan elimination by us

The proposed elimination approach reduces the quantum resources required.

- **Lower qubit and Toffoli count.** Removing the unused left-side updates reduces peak qubits by about 22% and Toffoli count by about 20% compared with J^+ [15].
- **Comparable depth.** Back-substitution adds only $r - 1$ Toffoli layers, about 0.3% for the evaluated parameter sets.

Table 2: Quantum resources for the Gauss-Jordan elimination compared with prior works [25,15].

Source	#Qubit*	# T^*	T -depth*	#Toffoli*	Toffoli-depth*
P^+ [25]	—	—	—	$\frac{r(r-1)(5r+8)}{6}$	$r^2 + 2r - 1$
J^+ [15]	$\frac{9r^2 - 11r + 6}{2}$	$\frac{7r(5r^2 + 3r - 8)}{6}$	$2r^2 + 2r - 4$	$\frac{r(5r^2 + 3r - 8)}{6}$	$\frac{r^2 + r - 2}{2}$
This work✳	$\frac{7r^2 - 5r + 4}{2}$ (22%)	$\frac{7r(2r^2 + 3r - 5)}{3}$ (20%)	$2r^2 + 6r - 8$ (-0.3%)	$\frac{r(2r^2 + 3r - 5)}{3}$ (20%/20%)	$\frac{r^2 + 3r - 4}{2}$ (50%/-0.3%)

* and *: Improvement percentages relative to J^+ [15] and P^+ [25]/ J^+ [15], respectively (a negative value denotes a degradation).

✳: Our work includes the added syndrome back-substitution.

Hamming weight check

The last stage of the oracle turns $H_S^{-1}s$ in the syndrome register into a marking decision.

- It computes the Hamming weight of this r -bit vector and flips the sign of the state when the weight equals t .
- The prior work computes this weight through $r - 1$ additions arranged in $\lceil \log_2 r \rceil$ layers.

This work accumulates the r bits of this vector with the quantum carry-save adder in parallel.

- This requires more ancillas than the adder tree.
- However, we reuse the clean ancilla qubits from the elimination phase, achieving a lower T-depth without additional qubits.

Logical resources for QISD

The two schemes exhibit different cost trade-offs: HQC has the larger matrix size, so its oracle uses more qubits, whereas its iteration count is smaller.

Table 3: Quantum resources for the ISD attack in base-2 logarithm.

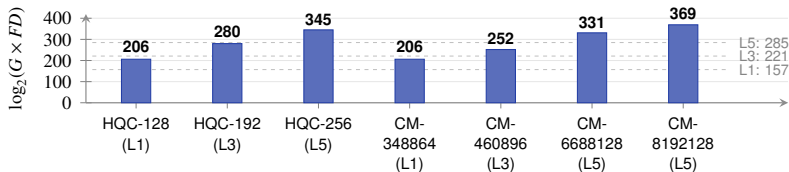
Scheme	Iterations	#Qubit	#Total gates (G)	T -depth (T_d)	Full-depth (FD)	$G \times FD^*$
HQC-128	59	30	105	89	101	206
HQC-192	93	32	142	125	138	280
HQC-256	123	34	175	156	170	345
Classic McEliece-348864	72	22	106	93	100	206
Classic McEliece-460896	93	23	129	116	123	252
Classic McEliece-6688128	131	24	168	155	163	331
Classic McEliece-8192128	150	24	187	174	182	369

*: The gate-depth cost metric used to evaluate the NIST post-quantum security levels.

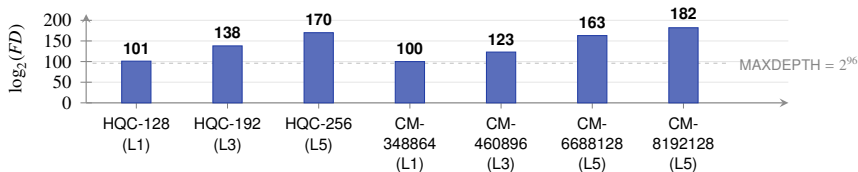
As a side note, compared with the space-optimized matrix-inversion approach [8], our elimination circuit lowers gates and depth, whereas the matrix-inversion approach lowers qubits.

NIST Post-Quantum Security

$G \times FD$ security cost



Full circuit depth



- Post-quantum security levels.** For every evaluated instance, $G \times FD$ exceeds the corresponding threshold: 2^{157} for level 1, 2^{221} for level 3, and 2^{285} for level 5.
- MAXDEPTH.** The full depth exceeds the MAXDEPTH upper bound of 2^{96} for every evaluated instance.

Fault-tolerant estimates: physical qubits and runtime

We estimate the physical-qubit count and the runtime.

- **Fault-tolerant model.** We adopt the surface code and Reed-Muller 15-to-1 magic-state distillation.
- **Hardware assumptions.** We assume a physical error rate $p_{\text{phys}} = 10^{-3}$, a per-gate error rate $p_g = 10^{-4}$, and cycle times of $1\mu\text{s}$, $10\mu\text{s}$, $100\mu\text{s}$, and 1ms .

The evaluated instances require at least 2^{35} physical qubits and 2^{53} years under this physical assumption.

Table 4: Physical resource estimates for QISD using surface codes (in base-2 logarithm).

Scheme	#Physical qubits	Runtime for code cycle time			
		$1\mu\text{s}$	$10\mu\text{s}$	$100\mu\text{s}$	1ms
HQC-128	43	53	57	60	63
HQC-192	46	90	93	96	100
HQC-256	49	121	124	128	131
Classic McEliece-348864	35	57	61	64	67
Classic McEliece-460896	37	81	84	87	91
Classic McEliece-6688128	39	120	123	127	130
Classic McEliece-8192128	39	139	142	146	149

Conclusion

This work optimizes the Gauss–Jordan elimination and Hamming-weight check in the QISD oracle, reducing quantum costs for HQC and Classic McEliece.

- Omit redundant updates and recover the result vector by back-substitution.
- About 22% fewer qubits than J+ and 20% fewer Toffoli gates than J+ and P+, with comparable Toffoli depth to J+.
- For HQC and Classic McEliece, $G \times FD$ exceeds the corresponding NIST security threshold, and FD exceeds the MAXDEPTH bound.
- The required physical qubits and runtimes are far beyond the capabilities of current quantum computers.

The following directions remain for future work.

- Considering more advanced ISD variants.
- Reassessing physical costs under lower-overhead fault-tolerant architectures.

Thanks and Questions

Thanks for your attention!